

Epic 7: Saving the Model

Introduction to Saving the Model

Saving a trained model is an important machine learning step.

It avoids retraining the model every time.

The trained model is stored on disk.

Python libraries like Pickle and Joblib are commonly used.

Saved models can be reused anytime.

This saves computation time.

It improves deployment efficiency.

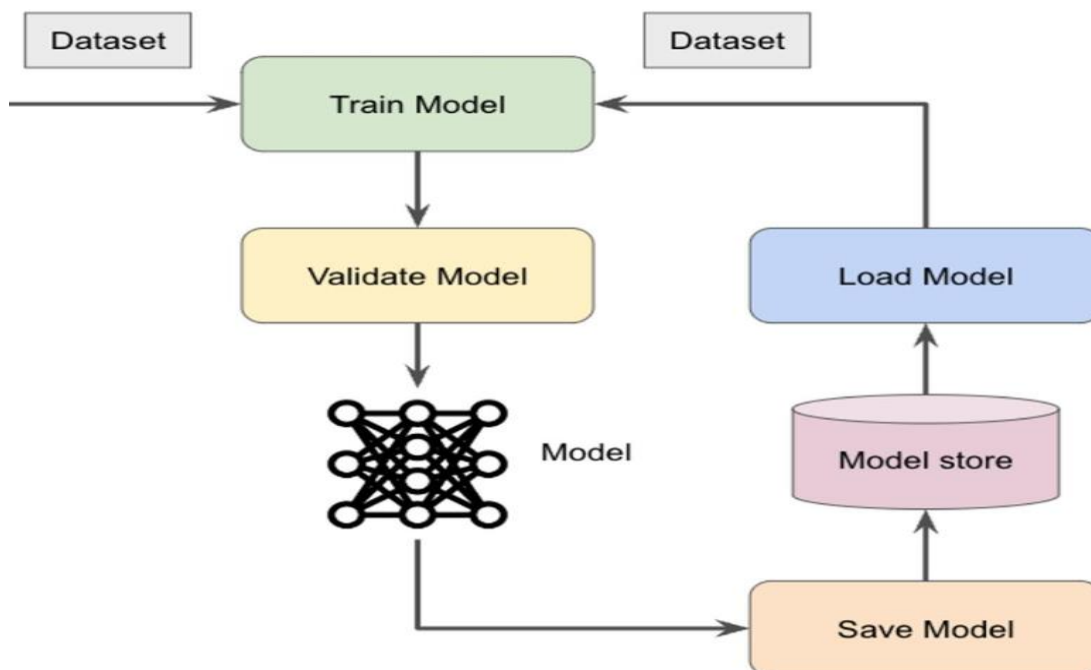
For the HDI project, the trained Linear Regression model can be stored.

The model file can be shared with other applications.

Saving ensures consistency in predictions.

It supports real-world deployment.

Model persistence is an essential practice.



Methods for Saving the Model

Joblib is preferred for large machine learning models.

Pickle is another popular serialization library.

The trained model is written to a file.

The file extension is usually .pkl or .joblib.

Loading the model is simple using the same library.

Saved models preserve learned parameters.

They eliminate repeated training.

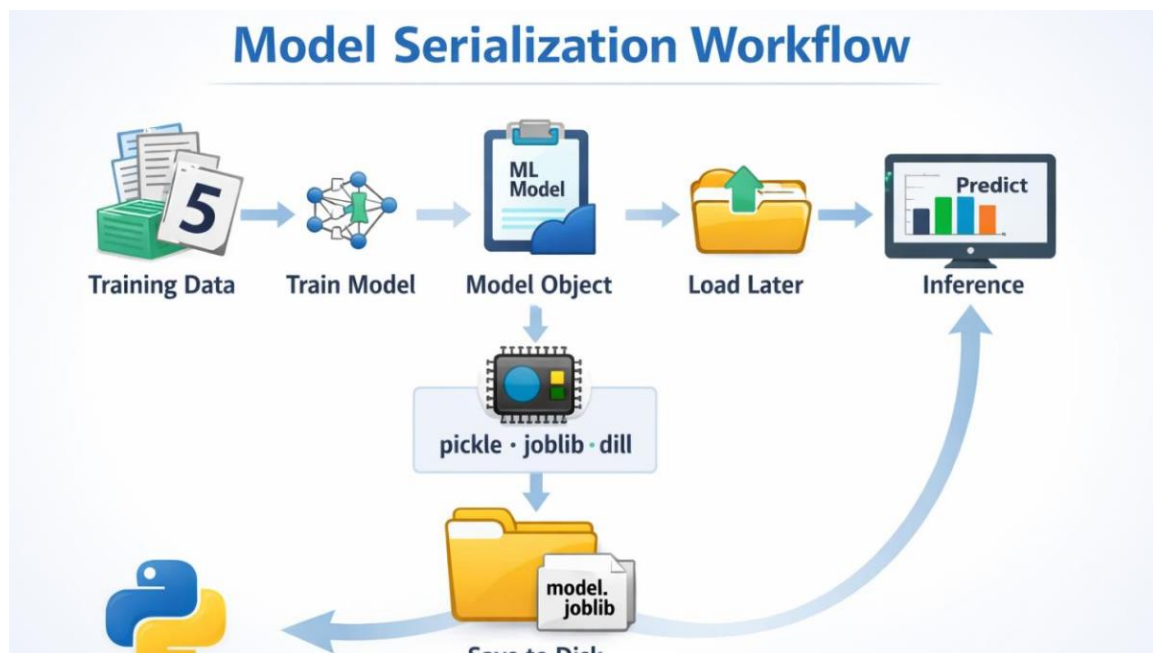
Version control helps manage updated models.

Model files should be stored securely.

Documentation should mention the save location.

Testing the saved model is recommended.

These methods simplify deployment.



Loading and Using the Saved Model

The saved model is loaded into memory.

New input data is passed to the loaded model.

The model predicts outputs instantly.

No retraining is required.

This improves application performance.

The prediction process remains consistent.

Saved models are used in web and desktop applications.

They support scalable AI systems.

Proper storage prevents data loss.

Model reuse reduces development time.

The HDI prediction system benefits from model persistence.

This completes the model saving process.

